Adding, Subtracting, and Multiplying Complex Numbers Answer Key

Algebra 2

Quick Answers

1. $7 + 3i$
2. $5 - 7i$
3. $21 + 12i$
4. $13 + 13i$
5. -1
6. $3 + 3i$
7. 29
8. $7 - 24i$
9. $1 - i$
10. $3 + 2i$, $3 - 2i$
11. $9 + 7i$
12. $x^2 - 10x + 26$, $5 + i$, $5 - i$

Curriculum

Level: Algebra 2
HSN-CN.A.1–A.3 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 11*
HSN-CN.C.7–C.9 – *problems 10, 12*
Difficulty 1–5 of 5 across 13 tagged checks.

Common wrong answers

7. *If they got 21: treated i squared as +1, so the last product read 25 minus 4 instead of 25 plus 4*

1. Simplify $(3 + 5i) + (4 - 2i)$.

Real parts combine with real parts, imaginary parts with imaginary parts — i behaves exactly like a variable here.
Real: $3 + 4 = 7$. Imaginary: $5i - 2i = 3i$.
$\boxed{7 + 3i}$

2. Simplify $(8 - i) - (3 + 6i)$.

Distribute the minus sign over both parts of the second number before combining:

$$8 - i - 3 - 6i$$

Real: $8 - 3 = 5$. Imaginary: $-i - 6i = -7i$.
$\boxed{5 - 7i}$
Grading note: $5 + 5i$ means only the real part of the second number was negated.

3. Multiply $3i(4-7i)$.
Distribute, then use $i^2 = -1$:

$$3i \cdot 4 - 3i \cdot 7i = 12i - 21i^2 = 12i - 21(-1)$$

The i^2 term becomes a real number, so it moves to the front.
$\boxed{21 + 12i}$

4. Multiply $(2+3i)(5-i)$.
FOIL first, replace i^2 second:

$$10 - 2i + 15i - 3i^2 = 10 + 13i - 3(-1) = 10 + 13i + 3$$

$\boxed{13 + 13i}$

5. Simplify i^{18}.
Powers of i repeat every four steps, so divide the exponent by 4 and keep the remainder: $18 = 4 \cdot 4 + 2$, remainder 2.
So $i^{18} = (i^4)^4 \cdot i^2 = 1^4 \cdot i^2 = i^2$.
$\boxed{-1}$

6. Simplify $(-4+7i) + (9-7i) - (2-3i)$.
Distribute the minus over the third number, then collect:

$$-4 + 7i + 9 - 7i - 2 + 3i$$

Real: $-4 + 9 - 2 = 3$. Imaginary: $7i - 7i + 3i = 3i$.
The imaginary parts of the first two numbers are additive inverses, so they cancel completely; the whole imaginary part of the answer comes from the third number.
$\boxed{3 + 3i}$

7. Multiply $(5+2i)(5-2i)$.
These are conjugates, so the outer and inner products cancel:

$$25 - 10i + 10i - 4i^2 = 25 - 4i^2 = 25 - 4(-1) = 25 + 4$$

The result is real because the $-4i^2$ term turns into $+4$: a complex number times its conjugate is always the real number $a^2 + b^2$.
$\boxed{29}$

Grading note: 21 is the diagnostic wrong answer — it means i^2 was treated as $+1$ (or the $-4i^2$ was simply read as -4). Every product problem on this sheet turns on that one substitution.

8. Expand $(4-3i)^2$.

Squaring a binomial is multiplying it by itself, so there are four products, not two:

$$(4-3i)(4-3i) = 16 - 12i - 12i + 9i^2 = 16 - 24i + 9(-1)$$

$\boxed{7-24i}$

Grading note: 25 means the student squared each term separately and dropped the middle. $16+9$ is not $(4-3i)^2$; it is the square of the modulus.

9. Simplify $i^{43} + i^{20}$.

Take each exponent mod 4.
$43 = 4 \cdot 10 + 3$, so $i^{43} = i^3 = -i$.
$20 = 4 \cdot 5 + 0$, so $i^{20} = i^0 = 1$.
Adding: $-i + 1$, which in standard form is written real part first.
$\boxed{1-i}$

10. Solve $x^2 - 6x + 13 = 0$ over the complex numbers.

The discriminant is negative, which is exactly when the solutions are a conjugate pair:

$$\Delta = (-6)^2 - 4(1)(13) = 36 - 52 = -16$$

$$x = \frac{6 \pm \sqrt{-16}}{2} = \frac{6 \pm 4i}{2}$$

Divide both parts of the numerator by 2.
$\boxed{x = 3+2i \text{ or } x = 3-2i}$
Check by multiplying: $(3+2i)^2 - 6(3+2i) + 13 = 5 + 12i - 18 - 12i + 13 = 0$ ✓.

11. Multiply $(2+i)(3-2i)(1+i)$.

Do it in two stages, returning to standard form after each stage — carrying an unsimplified i^2 into the next product is where this problem goes wrong.
Stage 1: $(2+i)(3-2i) = 6 - 4i + 3i - 2i^2 = 6 - i + 2 = 8 - i$.
Stage 2: $(8-i)(1+i) = 8 + 8i - i - i^2 = 8 + 7i + 1$.
$\boxed{9+7i}$

12. Synthesis: expand $\big(x-(5+i)\big)\big(x-(5-i)\big)$ and state its zeros.

Group the real part with x and treat the pair as a difference of squares:

$$\big((x-5)-i\big)\big((x-5)+i\big) = (x-5)^2 - i^2 = (x-5)^2 + 1$$

Expand the square: $x^2 - 10x + 25 + 1$.
$\boxed{x^2 - 10x + 26}$
The factors were built from x minus each root, so the zeros are exactly the two numbers that were subtracted.
$\boxed{x = 5+i \text{ or } x = 5-i}$
The coefficients came out real because the two roots are conjugates: their sum $(5+i) + (5-i) = 10$ and their product $(5+i)(5-i) = 26$ are both real, and those are precisely the coefficients the expansion produces.